



Retail Sales Performance Analysis using SQL

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Calculate total Revenue for total sales

```
SELECT SUM(price) AS Total_revenue  
FROM sales  
WHERE order_status = 'Delivered';
```

	Total_revenue
▶	31150.00

Calculate sum of sales in Category Wise

```
SELECT product_category,  
       SUM(price) AS total_sales  
FROM sales  
WHERE order_status = 'Delivered'  
GROUP BY product_category  
ORDER BY total_sales DESC;
```

	product_category	total_sales
▶	Electronics	13500.00
	Home	6800.00
	Clothing	5700.00
	Grocery	2750.00
	Beauty	2400.00

Calculate Order Distribution by City

```
SELECT city,  
       COUNT(order_id) AS Total_orders  
FROM sales  
GROUP BY city  
ORDER BY total_orders DESC;
```

	city	Total_orders
▶	Delhi	2
	Mumbai	2
	Lucknow	2
	Ahmedabad	2
	Chennai	2
	Bangalore	1
	Jaipur	1
	Pune	1
	Surat	1
	Kochi	1

Calculate Average, Minimum, and Maximum Order Value

```
SELECT AVG(price) AS Average_order_value,  
MIN(price) AS Lowest_order_value,  
MAX(price) AS Highest_order_value  
FROM sales  
WHERE order_status = 'Delivered';
```

	Average_order_value	Lowest_order_value	Highest_order_value
▶	2396.153846	600.00	6800.00

Calculate Highest Orders By Customer_name

```
SELECT order_id, customer_name, price
FROM sales
ORDER BY price DESC
LIMIT 5;
```

	order_id	customer_name	price
▶	12	Neha Verma	6800.00
	5	Suresh Kumar	5500.00
	7	Vikas Gupta	4200.00
	15	Suresh Kumar	4000.00
	2	Neha Verma	3200.00
⬇	NULL	NULL	NULL

Calculate Ranking Customers by Order Value

```
SELECT customer_name,  
       price,  
       ROW_NUMBER() OVER(ORDER BY price DESC) AS row_num  
FROM sales;
```

	customer_name	price	Row_num
►	Neha Verma	6800.00	1
	Suresh Kumar	5500.00	2
	Vikas Gupta	4200.00	3
	Suresh Kumar	4000.00	4
	Neha Verma	3200.00	5
	Amit Sharma	2500.00	6
	Rohit Patel	1800.00	7
	Amit Sharma	1500.00	8
	Rahul Singh	1200.00	9
	Sneha Iyer	1200.00	10

Calculate Customer Ranking

```
SELECT customer_name,  
       price,  
       RANK() OVER(ORDER BY price DESC) AS rank_position  
FROM sales;
```

	customer_name	price	rank_position
►	Neha Verma	6800.00	1
	Suresh Kumar	5500.00	2
	Vikas Gupta	4200.00	3
	Suresh Kumar	4000.00	4
	Neha Verma	3200.00	5
	Amit Sharma	2500.00	6
	Rohit Patel	1800.00	7
	Amit Sharma	1500.00	8
	Rahul Singh	1200.00	9
	Sneha Iyer	1200.00	9

Calculate Running Total Per Revenue

```
SELECT order_date,  
       price,  
       SUM(price) OVER(ORDER BY order_date) AS Running_total  
FROM sales  
WHERE order_status='Delivered';
```

	order_date	price	Running_total
►	2024-01-05	2500.00	2500.00
	2024-01-06	3200.00	5700.00
	2024-01-07	1200.00	6900.00
	2024-01-09	5500.00	12400.00
	2024-01-10	700.00	13100.00
	2024-01-12	950.00	14050.00
	2024-01-13	1800.00	15850.00
	2024-01-14	1200.00	17050.00
	2024-02-01	1500.00	18550.00
	2024-02-03	6800.00	25350.00

Calculate Moving Average of Total Sales

```
SELECT order_id,  
       order_date,  
       price,  
       AVG(price) OVER(  
         ORDER BY order_date  
         ROWS BETWEEN 2 PRECEDING AND CURRENT ROW  
       ) AS moving_average  
FROM sales;
```

	order_id	order_date	price	moving_average
▶	1	2024-01-05	2500.00	2500.000000
	2	2024-01-06	3200.00	2850.000000
	3	2024-01-07	1200.00	2300.000000
	4	2024-01-08	850.00	1750.000000
	5	2024-01-09	5500.00	2516.666667
	6	2024-01-10	700.00	2350.000000
	7	2024-01-11	4200.00	2350.000000
	8	2024-01-12	950.00	1950.000000
	9	2024-01-13	1800.00	2316.666667
	10	2024-01-14	1200.00	1316.666667

Calculate CTEs

```
WITH customer_sales AS (  
  SELECT customer_name,  
         SUM(price) AS Total_spent  
  FROM sales  
  WHERE order_status='Delivered'  
  GROUP BY customer_name  
)  
SELECT *  
FROM customer_sales  
ORDER BY total_spent DESC;
```

	customer_name	Total_spent
▶	Neha Verma	10000.00
	Suresh Kumar	9500.00
	Amit Sharma	4000.00
	Rahul Singh	1800.00
	Rohit Patel	1800.00
	Sneha Iyer	1200.00
	Priya Mehta	1200.00
	Kavita Joshi	950.00
	Anjali Rao	700.00

Calculate Combine Sales and Customer Details

```
SELECT s.order_id, s.customer_name, s.city,  
s.product_category, s.price, c.email,  
c.phone  
FROM sales s  
INNER JOIN customers1 c  
ON s.customer_id = c.customer_id;
```

	order_id	customer_name	city	product_category	price	email	phone
►	1	Amit Sharma	Delhi	Electronics	2500.00	amit@gmail.com	9876543210
	2	Neha Verma	Mumbai	Clothing	3200.00	neha@gmail.com	9876543211
	3	Rahul Singh	Lucknow	Grocery	1200.00	rahul@gmail.com	9876543212
	4	Priya Mehta	Ahmedabad	Beauty	850.00	priya@gmail.com	9876543213
	5	Suresh Kumar	Chennai	Electronics	5500.00	suresh@gmail.com	9876543214
	6	Anjali Rao	Bangalore	Clothing	700.00	anjali@gmail.com	9876543215
	7	Vikas Gupta	Jaipur	Home	4200.00	vikas@gmail.com	9876543216
	8	Kavita Joshi	Pune	Grocery	950.00	kavita@gmail.com	9876543217
	10	Sneha Iyer	Kochi	Beauty	1200.00	sneha@gmail.com	9876543219
	11	Amit Sharma	Delhi	Electronics	1500.00	amit@gmail.com	9876543210

Find All Sales Even if Customer Info is Missing

```
SELECT s.order_id, s.customer_name,  
s.product_category, s.price, c.email  
FROM sales s  
LEFT JOIN customers1 c  
ON s.customer_id = c.customer_id;
```

	order_id	customer_name	product_category	price	email
▶	1	Amit Sharma	Electronics	2500.00	amit@gmail.com
	2	Neha Verma	Clothing	3200.00	neha@gmail.com
	3	Rahul Singh	Grocery	1200.00	rahul@gmail.com
	4	Priya Mehta	Beauty	850.00	priya@gmail.com
	5	Suresh Kumar	Electronics	5500.00	suresh@gmail.com
	6	Anjali Rao	Clothing	700.00	anjali@gmail.com
	7	Vikas Gupta	Home	4200.00	vikas@gmail.com
	8	Kavita Joshi	Grocery	950.00	kavita@gmail.com
	9	Rohit Patel	Clothing	1800.00	NULL
	10	Sneha Iyer	Beauty	1200.00	sneha@gmail.com

Calculate Sales Orders with Customer Details

```
SELECT s.order_id, s.customer_name,  
       c.email, c.phone, s.product_category  
FROM customers1 c  
RIGHT JOIN Sales s  
ON c.customer_id = s.customer_id;
```

	order_id	customer_name	email	phone	product_category
▶	1	Amit Sharma	amit@gmail.com	9876543210	Electronics
	2	Neha Verma	neha@gmail.com	9876543211	Clothing
	3	Rahul Singh	rahul@gmail.com	9876543212	Grocery
	4	Priya Mehta	priya@gmail.com	9876543213	Beauty
	5	Suresh Kumar	suresh@gmail.com	9876543214	Electronics
	6	Anjali Rao	anjali@gmail.com	9876543215	Clothing
	7	Vikas Gupta	vikas@gmail.com	9876543216	Home
	8	Kavita Joshi	kavita@gmail.com	9876543217	Grocery
	9	Rohit Patel	NULL	NULL	Clothing
	10	Sneha Iyer	sneha@gmail.com	9876543219	Beauty

Complete Customer and Sales Data Analysis

```
SELECT c.customer_id, c.customer_name, c.email,  
       s.order_id, s.product_category, s.order_status  
FROM customers1 c  
LEFT JOIN Sales s  
ON c.customer_id = s.customer_id  
UNION  
SELECT c.customer_id, c.customer_name,  
       c.email, s.order_id, s.product_category,  
       s.order_status  
FROM customers1 c  
RIGHT JOIN Sales s  
ON c.customer_id = s.customer_id;
```

customer_id	customer_name	email	order_id	product_category	order_status
1	Amit Sharma	amit@gmail.com	11	Electronics	Delivered
1	Amit Sharma	amit@gmail.com	1	Electronics	Delivered
2	Neha Verma	neha@gmail.com	12	Home	Delivered
2	Neha Verma	neha@gmail.com	2	Clothing	Delivered
3	Rahul Singh	rahul@gmail.com	13	Grocery	Delivered
3	Rahul Singh	rahul@gmail.com	3	Grocery	Delivered
4	Priya Mehta	priya@gmail.com	14	Beauty	Delivered
4	Priya Mehta	priya@gmail.com	4	Beauty	Cancelled
5	Suresh Kumar	suresh@gmail.com	15	Electronics	Delivered
5	Suresh Kumar	suresh@gmail.com	5	Electronics	Delivered



Thank You